

The Evolution of Environmental Transparency: A Longitudinal Analysis of Symbolic vs. Substantive Disclosure (2002–2016)

Course: Data Analysis and Visualisation

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Abstract

This study examines the evolution of environmental disclosure in CSR reports from 2002 to 2016, focusing on the balance between symbolic and substantive language. Using a rule-based keyword approach, it finds that although reporting volume increased, substantive disclosure did not improve, and symbolic language became more dominant. While GRI G3 (2006) and G4 (2013) enhanced the reporting structure, they did not increase accountability. Overall, environmental reporting remains more communication-driven than performance-driven.

Keywords: Environmental disclosure, CSR reporting, Symbolic vs. substantive disclosure, Greenwashing, GRI frameworks, Text analysis

1. Introduction

Over the past two decades, environmental disclosure has become central to corporate social responsibility (CSR) reporting, driven by growing concerns about climate change and corporate accountability (Michelon, Pilonato & Ricceri, 2015). However, concerns remain about the credibility of disclosures (Cho, Roberts & Patten, 2010). A key distinction is between symbolic disclosure, which emphasizes intentions, and substantive disclosure, which reflects measurable outcomes (Cho, Roberts & Patten, 2010). Firms may use reporting strategically, often highlighting commitments without real operational change, a practice associated with greenwashing (Delmas & Burbano, 2011; Lyon & Montgomery, 2015). The introduction of GRI frameworks (G3 in 2006 and G4 in 2013) improved reporting structure but raises questions about their impact on substantive disclosure (Michelon, Pilonato & Ricceri, 2015). This study examines how environmental disclosure evolved between 2002 and 2016, focusing on the balance between symbolic and substantive language and the role of GRI frameworks.

2. Literature Review

Environmental disclosure does not necessarily reflect actual environmental performance. Prior research shows that firms may use sustainability reporting strategically, engaging in greenwashing and emphasizing reputational benefits over real environmental action (Delmas &

Burbano, 2011; Lyon & Montgomery, 2015). The GRI framework has standardised sustainability reporting and encouraged firms to focus on material issues (Michelon, Pilonato & Ricceri, 2015). However, studies suggest that firms may still present information selectively within these frameworks (Cho, Roberts & Patten, 2010). Recent research applies text analysis methods to examine disclosure language, with Huq and Carling (2024) proposing a framework to identify accountable environmental reporting. Despite these developments, limited research has examined how symbolic and substantive disclosure evolves over time or whether GRI frameworks have improved reporting quality. This study addresses this gap by analyzing changes in environmental disclosure language over time.

3. Research Questions

This study investigates the evolution of environmental disclosure quality in corporate sustainability reports between 2002 and 2016, with particular focus on the balance between symbolic and substantive language and the role of GRI framework transitions in shaping that balance. The following research questions guide the analysis:

- **Main RQ:** How has the nature of environmental disclosure in industries evolved from 2002 to 2016 in terms of language and reporting quality?
- **Sub-Q 1:** To what extent has the balance between symbolic and substantive disclosure shifted within these reports over time?
- **Sub-Q 2:** Did the introduction of GRI G3 (2006) and GRI G4 (2013) frameworks act as a catalyst in shifting industries from symbolic to substantive reporting?

4. Methodology

This study uses a rule-based text mining approach to analyse environmental disclosure in CSR reports from 2002 to 2016. All text files and classification rules come from the provided dataset and the environmental keyword lexicon.

A. Data Collection and Preparation

CSR report text files were loaded from the provided dataset and matched to their metadata. Only English reports from 2002 to 2016 with at least ten usable sentences were kept. Empty, missing,

or corrupted files were removed. Reports were then assigned to one of three GRI periods: Pre-GRI G3 (2002 to 2005), GRI G3 (2006 to 2012), and GRI G4 (2013 to 2016).

B. Sentence Extraction and Classification Framework

The classification system uses four word banks. Environmental Trigger Keywords identify relevant topics such as emissions, energy, water, waste, and biodiversity. Type 1 Words signal environmental action (e.g. reduced, deployed). Type 2 Words confirm results (e.g. achieved, verified). Symbolic Words indicate future plans or intentions (e.g. aims, roadmap).

A decision tree then classifies each sentence. SUBSTANTIVE sentences contain an Environmental Keyword with both Type 1 and Type 2 words, or use a past participle construction. SYMBOLIC sentences contain an Environmental Keyword and a Symbolic Word but no result confirmation. MIXED sentences carry both substantive signals and symbolic markers. NEUTRAL ENV sentences contain an environmental disclosure that does not necessarily reflect actual environmental performance. Environmental keyword without any action or aspiration markers. NON-ENV sentences have no environmental keyword. Only substantive, symbolic, and mixed sentences are used for quality analysis.

C. Report-Level Scoring

For each report, the following are computed: total sentences, environmental sentences, counts of each sentence type, TOTAL CLASS (substantive plus symbolic plus mixed), environmental density (environmental sentences divided by total sentences), and percentage shares for each type (count divided by TOTAL CLASS multiplied by 100).

D. Aggregation and Statistical Analysis

Reporting changes are tracked by grouping sentences by year to compare symbolic language against substantive results. Mean density is used to fairly compare reports of different lengths. The analysis covers reporting volume (Figure 5.1), the shift from evidence to intent (Figure 5.2), and the impact of GRI framework updates (Figure 5.3).

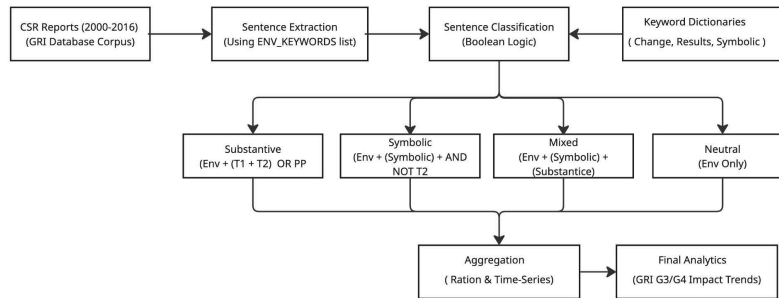


Figure 4.1

5. Findings

5.1 How Has Environmental Disclosure Changed Over Time?

Figure 5.1 tracks the mean density to fairly compare reports of varying lengths without skewing data. While density peaked at 34.38% in 2002 (n=4), it settled between 22% and 28% as the sample reached 317 reports by 2016. A 2014 peak of 33.20% aligns with early GRI G4 adoption, suggesting firms intensified their focus to meet new requirements during the transition. This downward trend shows that growth in reporting volume did not translate into higher environmental focus.

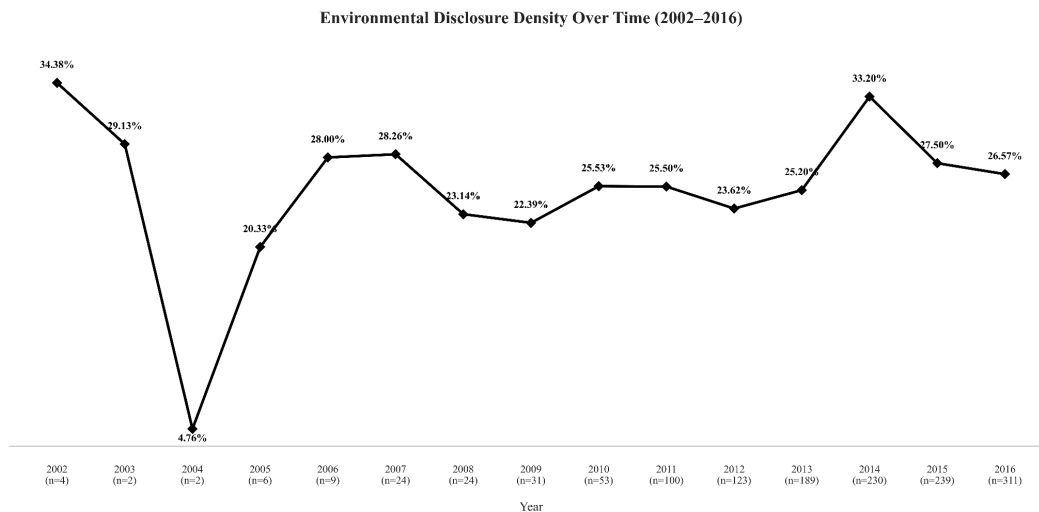


Figure 5.1

5.2 Has the Balance Between Symbolic and Substantive Shifted?

Figure 5.2 shows all three sentence types across the full period. In 2002, 43.3% of environmental sentences were substantive, and 55.9% were symbolic. By 2016, substantive dropped by 9.8 and

symbolic rose by 9.4 percentage points. Despite the small 2002 sample, the trend shows language shifting toward intentions rather than verified outcomes as reporting became mainstream. The "Mixed" category remained negligible (under 3%), indicating a clear divide between "talk" and "action".

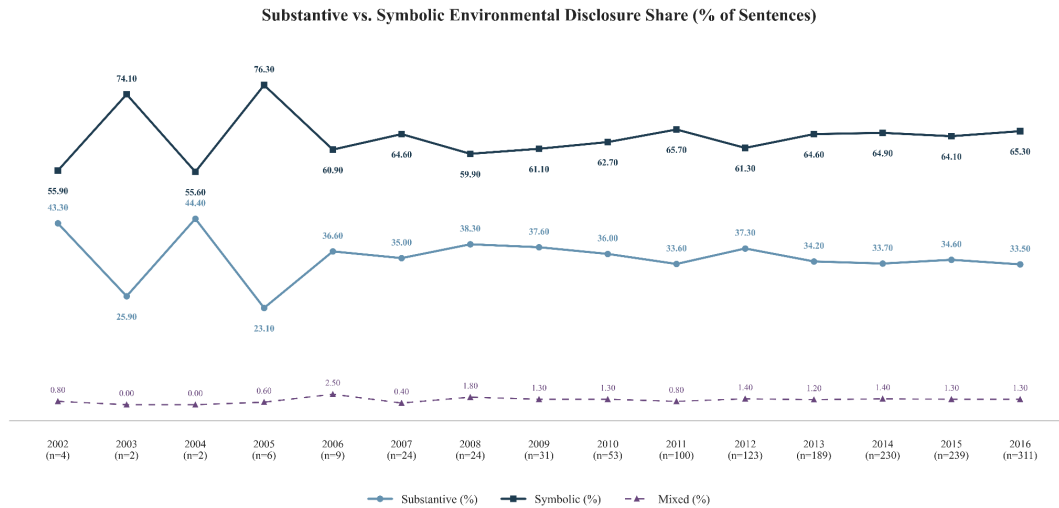


Figure 5.2

5.3 Did GRI G3 (2006) and GRI G4 (2013) Make Reporting More Accountable?

Figure 5.3 compares GRI eras where the pre-GRI G3 substantive share was 32.0%. While GRI G3 saw a minor rise to 35.9%, quality dropped back to 34.0% under GRI G4. Symbolic language remained dominant throughout, and the mixed category is excluded due to its small size. Ultimately, GRI frameworks increased reporting volume without significantly improving accountability.

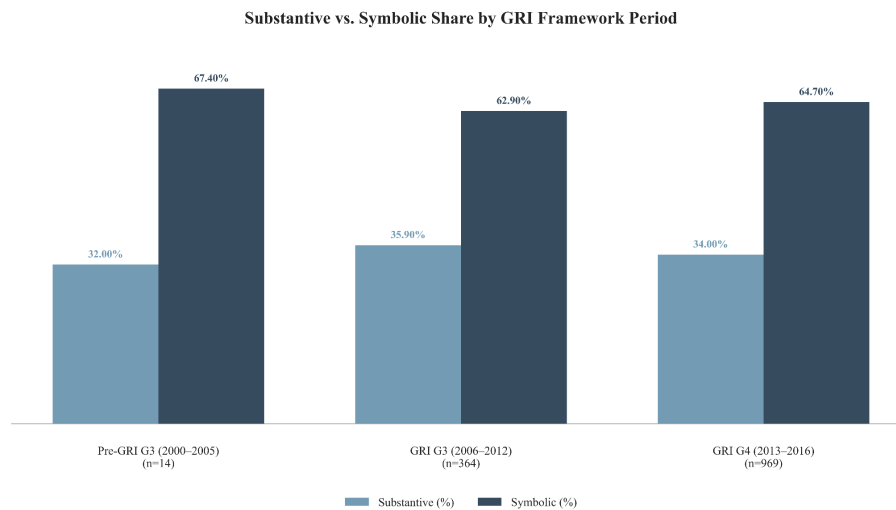


Figure 5.3

6. Conclusion

Based on the findings, between 2002 and 2016, companies increased their environmental reporting, but disclosures remained largely symbolic rather than substantive. Since the early reports (2002–2006) were few, the results should be interpreted cautiously. The introduction of GRI G3 (2006) and G4 (2013) improved structure and quantity but did not shift reporting toward verifiable environmental results. Overall, companies focus more on communication than on measurable environmental impact, highlighting implications for research and practice.

7. Limitations

Only English-language reports were analyzed, reducing the sample to 1,347 and introducing Western bias; the classification framework relies on keywords and sentence-level rules, so some environmental language may be missed, and the system does not handle negation, which may cause minor misclassifications. Converting PDFs to text may introduce noise, particularly in pre-2007 data. Finally, the dataset is uneven across time, sectors, and regions, with a concentration in 2011–2016, so findings should be seen as general trends rather than precise conclusions.

References

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